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ABSTRACT

Due to the emergence of generative artificial intelligence and its rapid growth, highly realistic synthetic images have been created on a large scale. This poses a big threat as it is becoming increasingly difficult to differentiate such images from real images. For the purpose of combating this problem, this project is based on automatically detecting AI-generated images using deep learning and providing a visual interpretation of the decision-making process. A Convolutional Neural Networks (CNN) is used to classify the input image as real or AI generated. It is trained and evaluated on a publicly available dataset containing both real and synthetic images. Though many such methods provide good classification results, they are often black-box models with poor interpretability. To deal with this problem, the proposed model incorporates features of Explainable Artificial Intelligence (XAI), which is accomplished using Grad-CAM. These Grad CAM models provide significant insights into the patterns and features learned by the CNN, thereby explaining the importance of the process. The combined approach of the detection model and the explanation model provides significant practical applicability to the detection system for the identification of fake images, verification of image content, and forensic analysis. This research aims to distinguish between genuine artworks and synthetically generated images using a benchmark dataset.

Keywords: *AI-Generated Image Detection, Deep Learning, CNN, Artificial Intelligence, Explainable AI, Grad-CAM, Synthetic Image Detection, Digital Image Analysis.*

CHAPTER 1

INTRODUCTI

ON

1.1 BACKGROUND

Artificial Intelligence, or AI, has been rapidly advancing in image generation over recent years, and in particular, through the development of what are known as diffusion models that can now convincingly and artificially create art that is indistinguishable from art generated by humans. This development brings issues of authenticity, potential misinformation, and challenges to digital trust.

To address this challenge, existing approaches have adopted deep learning techniques, including the deployment of Deep Learning approaches, particularly Convolutional Neural Networks that aim to distinguish between human-generated images and those created by current AI. Most of the existing approaches, however, work as black boxes, thus returning a classification decision without revealing the underlying justifications for such a decision.

In our project, our goal is to build a system that can distinguish between AI generated images and natural human images. We employ Error Level Analysis (ELA) as a preprocessing step to reveal faint compression artefacts embedded in the images. We then use a CNN to extract features from these processed images to classify between the two types of artwork.

To increase understanding of the model's decisions, Grad-CAM was used as an Explainable AI (XAI) technique to create visualizations of the areas that affected the model's score most.

In our work, we aim to present an accurate and explainable deep learning approach for detecting artificially generated images. It can be used for a variety of tasks in digital forensics and content verification.

1.2 PROBLEM STATEMENT

Generating high quality images has evolved as a rapidly developing field in recent years with Generative AI, resulting in some images being so convincing they often look like they were created by humans themselves, making it increasingly difficult for people to identify real and fake images.

Artificially generated images are everywhere – this is great for entertainment purposes such as movies and games. Yet there are some very serious and important aspects to consider like is this going to increase misinformation and/or fake news, who owns the rights to something generated by AI and will programs be able to recognize “fakes” to help aid digital crime scene

investigations in the near future? If we are to tackle all of these challenges, the first step is to have an adequate image forensic system that can successfully detect forged images.

Most state-of-the-art deep learning solutions currently utilising deep neural networks to classify AI generated images exist as so called black-boxes: meaning although they have high accuracy at image classification they lack any kind of understandability or explainability for their given results, which currently means that users do not have much trust in these models and as a result cannot apply them in practical real world scenarios.

For rigorous identification of AI-generated content, an accurate and explainable image detection system is needed; this requires an ability to visually explain its decisions. Using a suite of Explainable AI (XAI) techniques provides one way to increase interpretability, improve trust, and product a more reliable and practical tool for image forensics and content verification tasks.

1.3 LITERATURE SURVEY

In recent years, various approaches based on deep learning and feature-based methods have been applied to the problem of detecting AI-generated images.

Yavuz Selim Taspinar (2024) proposed a deep learning architecture for fake image detection based on the usage of pre-trained versions of popular CNN models with a technique known as transfer learning. This approach is very simple to implement, has very good feature extraction capabilities and, as a result, achieves very competitive performance results on the images in which it is trained. However, it has very low generalization capabilities to novel generative models and most importantly, it is opaque in terms of understanding of its decisions, merely working as a black-box.

In Bird and Lotfi (2023), CIFAKE videos and frames were introduced and addressed using a CNN-based binary classifier and backed up with Grad-CAM explanations, where the model performed well on classification as well as provided insights into the decisions. However, the model failed to generalise to generated images (from outside the training distribution) quite severely.

In Francesco Cozzolino, et al., Multi-Modal Detection (MMD) Detectors based on CLIP features combine strengths of Image Models and Language Models achieving state of the art accuracy levels and semantic understanding. Nevertheless, their computational complexity prevent their real-time deployments.

Zheng et al. (2024) present a patch-based CNN denocator that breaks semantic artifacts in images generated by AI models. The method is evaluated on several different generative models including GANs and diffusion models. The method is very powerful but requires a complicated training pipeline and high computational cost.

In Yang et al. (2025), the authors explored frequency-domain and texture-based features to detect deepfakes, AI-generated images. They report high accuracy but their approach requires a lot of preprocessing and does not robustly handle image compression.

In Yumlembam et al. (2025), a system was designed that combined several uncertainty measures for AI image detection. This approach provided an effective means of pruning out incorrect classifications and achieving improved performance on untested data. However, the resultant model increased in terms of complexity and requires the manual adjustment of multiple components.

Although existing approaches based on ELA achieved high accuracy in natural language processing tasks, they often failed to generalise to unseen models and, crucially, lacked interpretability. This project goal is to improve the both performance and interpretability of existing approaches by combining ELA-based preprocessing with a CNN text classifier and the Grad CAM technique for explainability.

1.4 SYSTEM REQUIREMENTS

1.4.1 Hardware Requirements

- **Processor:** A system equipped with an Intel i5 / AMD Ryzen 5
- **RAM:** Minimum 8 GB (16 GB recommended for improved performance)
- **Storage space:** At least 10 GB of available storage space
- **GPU:** GPU support such as Google Colab GPU (T4) for faster training
- **System Type:** 64-bit system

1.4.2 Software Requirements

- **Operating System:** using Windows / Linux / macOS
- **Programming Language:** using Python (version 3.8 or above)
- **Development Environment:** Google Colab
- **Libraries used:**
 - Tensor Flow / keras for deep learning
 - NumPy for numerical operations
 - OpenCV for image processing
 - Matplotlib for visualization
 - Scikit learn for data splitting & evaluation
- **Tools Used:**
 - Google Colab (model training with GPU)
 - Kaggle (dataset source)

1.5 SCOPE OF THE PROJECT

This project is exploring the possibility of building a computer vision system that can tell if an image was generated by a human or a computer. The system would use deep learning techniques for classifying images as either human or AI generated. It would integrate modern techniques in deep learning area with those in image forensics. The system would be designed to deal with the increasing problem of distinguishing between real and synthetic images.

This project uses a large dataset of over 25000 images from various artistic styles – both human and AI generated – and builds on the work of compressing images and utilising Error Level Analysis (ELA) to highlight subtle compression errors in the images. The preprocessed images are given as input into a Convolutional Neural Network to train the network to distinguish between compressed and uncompressed images for both human and AI images.

To increase the model's understanding, Explainable AI methods like Grad CAM have been integrated into the system to make the predictions more transparent and explainable, which is particularly relevant for real-world use cases and has huge potential to make models more trustworthy.

Our project also involves a comparison of approaches for detection such as vanilla CNN-based approaches and explainable approaches. We have made a simple GUI which can be used to upload images and get results along with explanations in real time.

The system's performance is measured using the accuracy and the validation performance. The system's ability to generalise to new images, created by AI, will also be evaluated. The goal of the system is to achieve high detection accuracy and show good generalization capabilities.

It is important to note that this project is focused on image classification. It is not dealing with video-based deepfake detection or any large-scale, real-time applications. Consequently, the results obtained may not generalise to images that have been heavily compressed or otherwise modified from their original form, even if found on the web and in public datasets.

This project forms a basis for future work on constructing robust and explainable AI-based image detection tools that can be used in variety of fields including digital forensics, media verification, and ethical AI.

1.6 MOTIVATION

The internet is awash with Images Generated by AI (Deepfakes, specifically) that are becoming increasingly realistic and far more prevalent; it is no longer sufficient for human eyes and brains to simply tell the difference between “genuine and real” versus “artificial and fake”, as new models such as diffusion-based models are able to generate images indistinguishable from human-generated ones.

Existing image detection techniques generally focus on visible features and simple patterns, which are insufficient for uncovering the often subtle artefacts that are generated by state of the art AI models. Most current deep learning-based image detection methods are black-box solutions that are able to achieve high accuracy but which provide little insight into how

decisions are made, thus leaving a large trust gap for applications such as digital forensic analysis, media verification, and content authenticity.

This challenge raises the bar to not only an accurate detection of deep learning generated images but also an accompanying explanation of the decisions made by the model. By incorporating Error Level Analysis (ELA) into an image forensics solution, hidden compression artifacts and inconsistencies can be revealed which in turn enable a deep learning model to learn features which are more meaningful than just looking at the images' appearance.

We also implemented Explainable AI methods like Grad CAM to allow for the visualization of regions the system focuses to make a decision and assesses whether the model is paying attention to informative or spurious features.

We combine techniques from deep learning, image forensics and explainability to develop a transparent, robust and accurate detection approach for DOS attacks.

CHAPTER 2

OBJECTIVE

S

We are working on a project that aims at building a deep learning-based system capable of classifying images with high accuracy and explainability. As generative models are getting better at generating human-looking images, there is an ever-increasing need for developing systems that can detect whether an image was generated by an AI or by a human.

The primary goal of this research involves using CNN to learn such complex visual, hierarchical representations that can distinguish between real images and images generated by a computer. Deep learning is used to automatically learn features of images and classify images at a depth within the network that manually extracting features from images cannot recognize.

In addition to the main objectives, we aim to utilise Error Level Analysis (ELA) as a preprocessing technique to reveal concealed or imperceptible characteristics of hidden data within images, with a view to improve detection of AI-generated content that looks realistic but diverges in structure from naturally generated images.

In order to reach the best detection results, we plan to implement a hybrid network that merges the RGB features with the features obtained from ELA in order to exploit semantic information and artefact-specific features characterizing different images and in order to improve the generalization capabilities.

To increase the trustworthiness of the proposed architecture for content classification and authentication, the goal of this part of the project is to integrate techniques for the explainability of AI (XAI), and in particular to use Score-CAM to generate visual explanations of deep learning models' decisions. Providing information about the parts of the image that account for the largest portion of the score for the assigned class increases the interpretability of the system. XAI will facilitate both digital forensics and content verification.

The project aims to also implement and test different models to recognize the most suitable one to use in the project. The implemented models include a hybrid CNN model, a transfer learning model based on EfficientNet, and a baseline CNN model to compare the results and to understand which method is the most effective for the project's goals.

We also aim to evaluate the system on evaluation metrics such as accuracy, validation performance and generalisation performance on images generated by new, never before seen generative models. This makes sure that the system is not only performing well on training data, but is also able to handle unseen data.

Our goal with this project is to create an accurate and scalable approach to finding images generated by AI, with usage in digital forensics, media authentication and more—to make sure that AI is developed and used ethically.

CHAPTER 3

METHODOLOGY

GY

3.1 DATASET PREPROCESSING AND FEATURE EXTRACTION

3.1.1 Dataset Description and Organization

In this project, we use the AI-ArtBench Dataset (Dataset) that contains a collection of 2 categories of art: human/physical and digitally produced by artificial intelligence (AI). This dataset is made up of 185,015 total images - 60,000 human generated images and 125,015 AI-generated images using the following models: Latent Diffusion and Standard Diffusion.

The AI-ArtBench Dataset contains ten different artistic styles (e.g. Impressionism, Realism, Surrealism, Renaissance, Ukiyo-e) in order to create a sufficient mix of visual patterns. A diverse dataset like this will allow the model to generalise across a number of different styles; otherwise, the model would overfit to an individual category/style.

The methodology to assess performance and generalisation is to structure the dataset:

- **Training Dataset:** Human + Latent Diffusion images
- **Validation Dataset:** Used for hyperparameter tuning
- **Testing Dataset:** Used for final evaluation

This separation ensures that the model is tested on **unseen generative techniques**, making the evaluation more realistic.

Dataset Split Distribution

Table 1.1 – Dataset Split Distribution

Dataset Split	Human Images	AI Images (LD)	Total Images
Training	~35000	~36000	~71000
Validation	~7500	~7800	~15300
Testing	~7500	~7800	~15300

Table 1.2 - Dataset Statistics

Parameter	Value
Dataset Name	AI-ArtBench
Total Images	185015
Human-Created Images (Total)	60000
AI-Generated Images	125015
└ Latent Diffusion	60000
└ Standard Diffusion	65015
Number of Classes	2(Human, AI)
Image Size	256*256*3
Artistic Styles	10 Categories
Training Data Split	70%
Validation Data Split	15%
Test Data Split	15%

3.1.2 Image Pre-processing

Image pre-processing plays a vital role in increasing the performance and ability of models in deep learning. In this project, multiple required preprocessing steps are performed to standardize the input data and enhance differentiating features required for finding AI generated images:

a) Image Resizing

All the images present in the dataset are resized to 256×256 pixels to make sure and set a uniform input size for the CNN model. This helps in reducing computational costs while holding important visual information.

b) Color Space Conversion

In this process, all the images are converted from BGR to RGB format, because OpenCV loads images in BGR format while deep learning models require RGB format. This makes sure correct color representation for feature extraction.

c) Pixel Normalization

In this process, value of each pixel is scaled from the range of 0-255 to 0-1. Normalizing the pixel values is important because it provides better stability for training the model and helps the model to learn fast during the training.

d) Error Level Analysis (ELA)

Error Level Analysis is an important preprocessing step that is used to mark the compression artifacts.

- The image is recompressed at **90% JPEG quality**
- The difference between original and compressed image is computed
- The difference is amplified to enhance visibility Observation:
- AI images → uniform patterns
- Human images → irregular patterns

ELA helps the model to detect micro details that are not visible to the human eye.

e) Final Input Preparation

After performing all pre-processing steps, the images are converted into array-formatted images of shape $(256 \times 256 \times 3)$ appropriate for suitable model input.

3.1.3 Data Splitting

To handle class imbalance, the dataset is split using stratified sampling into:

- **70% Training**
- **15% Validation**
- **15% Testing**

Stratification sampling ensures equal representation of both the label classes present in the dataset. (AI label and Human label).

3.2 DATA PIPELINE

The tf.data API is used for the Data Pipeline in order to create a TensorFlow-based solution that is fast to load data through the pipeline.

The processing stages of the pipeline include:

- Dynamic image loading from file paths
- Preprocessing (ELA, resizing, normalization)
- Batch formation (batch size = 32)
- Prefetching for improved training speed

This pipeline ensures efficient utilization of computational resources and faster training.

3.3 SYSTEM ARCHITECTURE

The proposed system is a multi-stage pipeline utilizing a combination of AI-based image forensics techniques to detect digital artwork that was created using AI. It is comprised of three key components; Preprocessing, Deep Learning Model Processing, and Explainable AI Processing to ensure that results are accurate and easily interpretable.

3.3.1 Input Layer

The user of the system will enter an input image as part of the systems interface through either a user-interface or dataset. Examples of the input may be any format of digital artwork; JPEG or PNG, etc. This allows the user to utilize both datasets and global internet images as input to the system.

3.3.2 Preprocessing Layer

This stage processes the input image to achieve a standardized and enhanced feature set:

- Resizing: All images are resized to a fixed size (256×256)
- Normalization: All pixel values will be normalized to the range of $[0, 1]$

- Error Level Analysis (ELA): This method will generate a secondary image for analysis

The Result of the ELA process helps to detect AI generated images through producing a representation of inconsistencies due to compression between two images. Therefore, this stage provides for two parallel processed images to enter into the next stage of the pipeline:

- An RGB image representation of the original image.
- An ELA representation of the original.

3.3.3 Feature Extraction Layer

The feature extraction layer facilitates the extraction of descriptive features that can be used for model training based on both inputs in parallel.

- Features used to train the model for RGB images as well as high-level semantic features such as Shapes, Textures and Patterns will be drawn from a pre-trained EfficientNet.
- Low-level artifacts and inconsistencies associated with image compression will be drawn from an ELA image using a custom CNN.

With this approach to architecture, the system will be able to learn two distinct types of information about an image.

- Visual content
- Forensic content

3.3.4 Feature Fusion Layer

The feature fusion layer integrates the outputs of the features extracted from the RGB and ELA images through the use of concatenation. The result of this is a representation that contains both semantic the forensic features. The final features used for classification will pass through the following processing functions:

- Use of batch normalization to stabilize the training of the model.
- Use of fully connected (dense) layers to learn complex relationships in the data.
- Use of dropout to reduce the risk of overfitting the model.

The combined nature of the different types of features (i.e., semantic and forensics) will improve the classification accuracy of the model once completed.

3.3.5 Classification Layer

To obtain a final “output” value between 0 and 1, the combined features of both inputs are fed to a sigmoid activation level.

- If the output is very low (near 0), then it can be concluded that this image is created by a human.
- If the output is very high (near 1) then it can be determined that this image was created by an artificial intelligence (AI).

Binary layer classification is performed by this layer based on learnt patterns during model training.

3.3.6 Explainability Layer (Grad-CAM)

The explainability of the system is achieved through the use of Grad-CAM as follows:

- Using extracted feature maps from the last convolutional layer.
- Each of the feature maps acts as a mask on the input image.
- The model’s prediction score is determined for each of the masked input image(s).
- By taking a weighted combination of all of the feature maps, a heatmap is generated to show which regions within the input image contributed to generating the final predicted class.

Ultimately this heatmap shows the location(s) in the input image that contributed to the specific predicted class (AI vs Human).

3.3.7 Output Layer

In summary, when an image has been processed/analysed through the Explainable AI system, the following information will be available at the output of the system:

- Predicted class (AI or Human)
- Confidence score
- ELA image
- Heatmap visualization
- Overlay of heatmap on original image

The output information generated by the systems will all be presented via a GUI and will establish user-friendly interactivity.

3.3.8 System Flow Summary

The entire workflow process of the Explainable AI system can be summarized using the flow diagram shown previously:

Input Image → Preprocessing (Edge Level Analysis + Normalising) → Feature Extraction (RGB Feature + ELA Feature Extraction) → Fused Features → Predict Class → Score-CAM → Output Functionality (View Output Results via user graphics interface)

3.3.9 Architecture Diagram

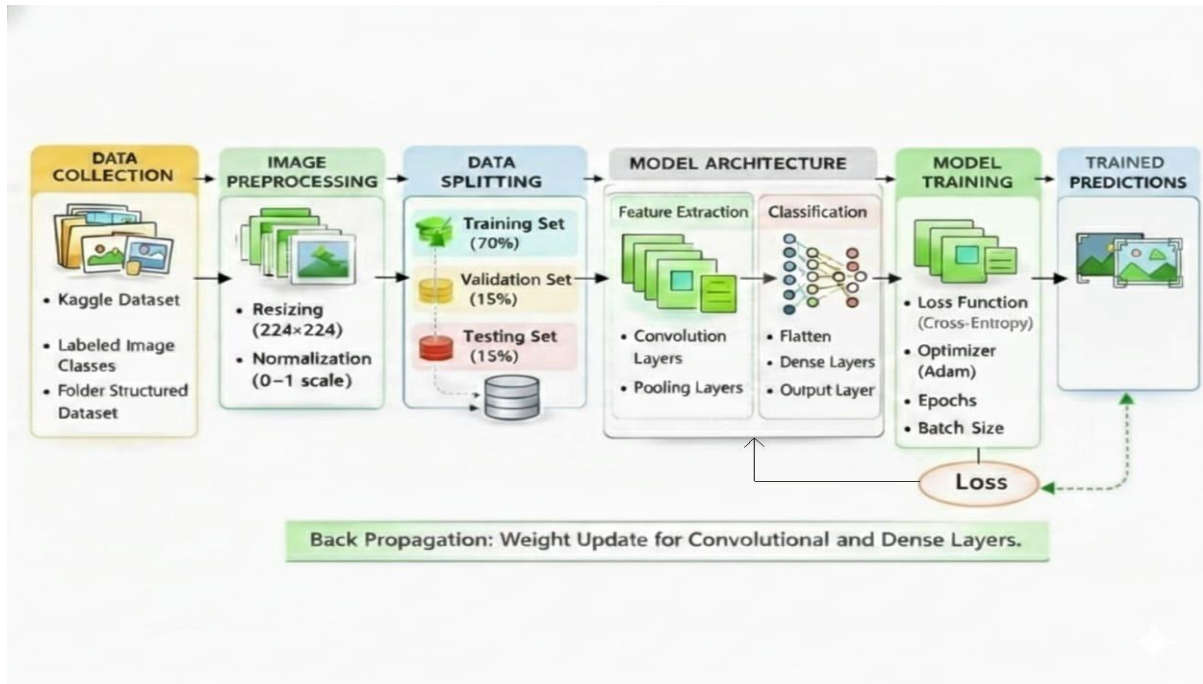


Fig.3.1. Architecture Diagram

3.4 SYNOPSIS OF THE ALGORITHMS USED

3.4.1 Convolutional Neural Networks (CNN)

CNN is a deep learning model designed to process data with a grid like topology such as images. CNN is used in the project because of image classification.

Working Principle:

A CNN processes images with processing filters through convolution operations to detect patterns within an image. The architecture consists of:

- Convolution Layers → Extract spatial characteristics of an image such as edge or texture
- Activation Functions (ReLU) → Introduce non-linear characteristics in the neural network

- Pooling Layers → Reduces the image's spatial resolution while retaining needed characteristics
- Fully Connected Layers → Classification of the object in an image

Logic in this Project:

CNN is used to:

- Learn hierarchical features from ELA images
- Capture compression artifacts
- Distinguish between structured (AI) and irregular (human) patterns

3.4.2 EfficientNet (Transfer Learning Model)

EfficientNet is the latest CNN architecture optimally used by scaling depth, width and resolution.

Working Principle:

EfficientNet uses compound scaling:

- Depth → Total number of layers
- Width → Total number of channels
- Resolution → Size of the Input image

The compound scaling of EfficientNet increases computer vision performance without creating a significant increase in the computational cost of running a neural network.

Logic in this Project:

- EfficientNetB0 is used in the hybrid model for RGB feature extraction
- EfficientNetB4 is used as a standalone model for benchmarking

These models capture:

- High-level semantic features
- Artistic patterns and structures

3.4.3 Feature Fusion Technique

The feature fusion combines multiple feature representations into one vector.

Working Principle:

- Features from RGB and ELA branches are concatenated

- Dense layers learn relationships between combined features

Logic in this Project:

- The RGB features capture the visual semantics of the image.
- The ELA features capture the compression artifacts of the image.

Combining both together can improve classification accuracy and robustness.

3.4.4 Grad-CAM (Explainable AI)

Grad-CAM is technique that produces visual heatmaps, highlighting the crucial image regions a Convolutional Neural Network (CNN) uses to make predictions.

Working Principle:

1. Extract feature maps from the last convolution layer
2. Normalize and resize each feature map
3. Use each feature map as a mask on the input image
4. Compute prediction scores for masked images
5. Generate heatmap using weighted combination

Mathematical Representation:

Heatmap is computed as:

$$N \text{ Heatmap} = \sum_{i=1} S_i F_i$$

Where:

- S_1 = prediction score for masked image
- F_1 = feature map

Logic in this Project:

- AI images → show large concentrated regions
- Human images → show scattered patterns

Score-CAM provides:

- Better stability than Grad-CAM
- More reliable explanations

3.4.5 Binary Classification using Sigmoid

The final classification is performed using a sigmoid activation function.

Working Principle:

$$\sigma(x) = 1/(1+e^{-x})$$

- Outputs probability between 0 and 1
- Threshold used to classify AI vs Human

Logic in this Project:

- Score > 0.5 → AI-generated
- Score < 0.5 → Human-created

3.5 MODEL TRAINING CONFIGURATION

In this project, we implemented three different models, Hybrid CNN (RGB + ELA), EfficientNet-B4 and Baseline CNN. We then trained and evaluated the three models using a pipeline. The dataset is divided into three parts: training, validation and test sets. We used stratified split, meaning that in each set, the class distribution was maintained. Hybrid CNN and EfficientNet models were trained using the complete dataset, while Baseline CNN was trained on a much smaller subset for comparison purposes.

Table 3.4 Model Training Configuration

Parameter	Hybrid CNN (RGB + ELA)	EfficientNet-B4	Baseline CNN
Dataset Split	80% Train / 20% Val	80% Train / 20% Val	80% Train / 20% Val
Input Size	224 × 224 × 3 (RGB + ELA)	224 × 224 × 3	224 × 224 × 3
Optimization Algorithm	Adam Optimizer	Adam Optimizer	Adam Optimizer
Learning Rate	1e-4 → 1e-5 (fine-tune)	1e-4 → 1e-5 (fine-tune)	0.001
Loss Function	Binary Crossentropy	CrossEntropyLoss	Binary Crossentropy

Pretrained Weights	EfficientNet-B0 (frozen → unfrozen)	Yes (EffNet-B4)	No
Regularization	Dropout (0.5), BatchNorm	Implicit (pretrained)	EarlyStopping, LR Reduce

3.6 EVALUATION METRICS

To assess the proposed models, a thorough set of evaluation metrics are utilized. Since the classification process is binary, i.e., images are classified between AI-generated and human-created images, the chosen metrics reflect different aspects of the classification performance metrics. The performance of the proposed models is assessed using the test images to verify their ability to generalize.

Accuracy:

Accuracy gives the proportion of correctly classified predictions out of the total number of predictions. It provides an overall measure of model performance.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

Precision tells you the number of positive cases predicted correctly as a percentage of the total positive cases predicted. A highest precision value will indicate the lowest number of false predictions for positive cases.

$$Precision = \frac{TP}{TP + FP}$$

Recall(Sensitivity):

Recall (or sensitivity) is used to measure the total number of actual positive samples that the model has predicted correctly. Thus, it is a good indication of the ability of the model to find all correct samples.

$$Recall = \frac{TP}{TP + FN}$$

Specificity:

Specificity indicates how many cases were predicted by the model correctly as negative. A high specificity value indicates that the model has verified that some cases were not true positives before making its predictions.

$$Specificity = \frac{TN}{TN + FP}$$

F1-Score:

The F1-score is a derived number based on the precision and recall numbers. Since F1-score is a harmonic mean of both numbers, it can be seen as a value that provides the average performance of your model when a high score on false positives is balanced against a low score on false negatives.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

ROC-AUC:

ROC – AUC measures the ability of a model to discriminate between the image possibilities from AI vs. Human by measuring the model's confidence levels over all the possible classification thresholds for each observation. The larger the AUC, the higher the performance in discriminating between the AI images & Human images.

Confusion Matrix:

The confusion matrix is a 2×2 matrix that is used to visualize classification performance of the model. Rows represent actual classes, while columns represent predicted classes. The diagonal elements indicate correct predictions, while off-diagonal elements represent misclassifications. It provides detailed insight into how well the model distinguishes between AI-generated and human images.

Where TP = True Positive, TN = True Negative, FP = False Positive, and FN = False Negative.

CHAPTER 4

RESULTS &

DISCUSSION

In this section, experimental outcomes of the three developed models are briefed. The results of Hybrid CNN model (EfficientNet-B0 with ELA feature fusion), EfficientNet-B4 model and baseline CNN model are evaluated based on accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix plots for classification task of distinguishing images generated by AI and images created by humans. Training and validation results obtained from experimental runs on images in the dataset are reported alongside the results obtained while testing the models on additional images samples. A comparative study is performed to discuss the effectiveness and limitations of developed models and to identify which model performed best in images generated by AI.

4.1 HYBRID CNN MODEL RESULTS

4.1.1 Training Performance



Fig. 4.1. Training and Test Accuracy Curves – Hybrid CNN Model

The Hybrid CNN is trained in two phases: in the initial training phase and in the fine-tuning phase. In the initial training phase, the network was trained for 6 epochs in which the training

accuracy and the validation accuracy both increased. The training accuracy increased from 70.5% up to a maximum of 88.0%, while the validation accuracy increased from 65.7% up to a maximum of 90.1%.

During evaluation of the performance of the model with all learning rates unfrozen for the fine-tuning phase, I also managed to improve the feature learning and overall validation accuracy of the upper layers. The top layers of EfficientNet-B0 were unfreeze and with some tuning the model reached an accuracy of 91.72%.

There is only a little difference between the training error and the cross-validation error; this does not indicate significant overfitting. Accurate on test: 92.43%.

4.1.2 Visualization of Evaluation Metrics

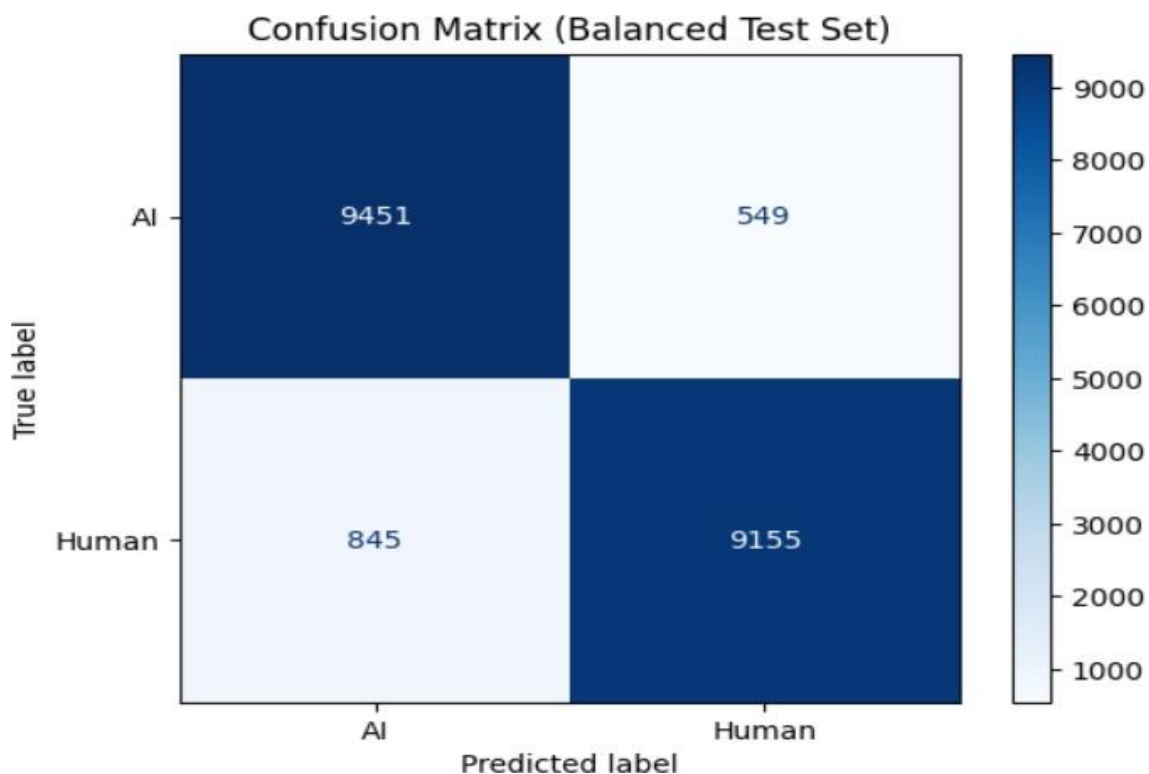


Fig. 4.2. Confusion Matrix – Hybrid CNN Model

From the confusion matrix, it can be observed that:

- The model correctly classifies a large number of both AI and human images.
- The number of misclassifications is relatively low compared to correct predictions.
- Slight imbalance exists due to the dataset distribution, where AI samples are more dominant in the test set.

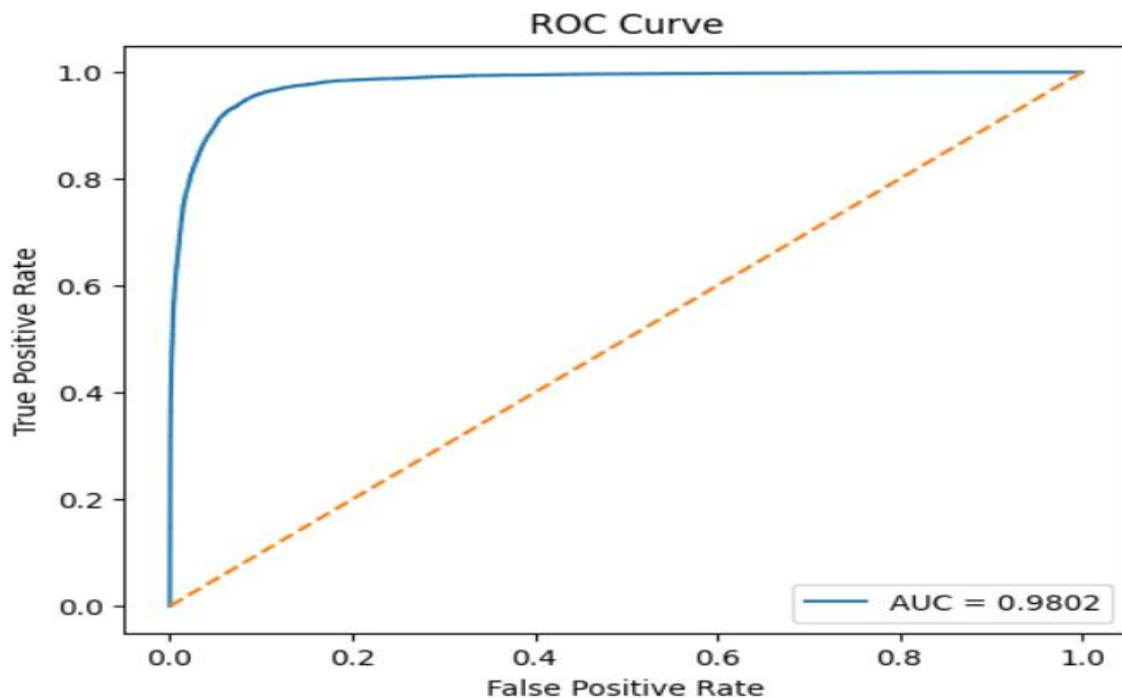


Fig. 4.3. ROC Curve – Hybrid CNN Model

Table 4.1 summarizes the general performance of the Hybrid CNN model:

Table 4.1 Hybrid CNN – Performance Metrics

Class	precision	Recall	F1-Score	Specificity	ROC-AUC
AI	0.9572	0.9441	0.9506	0.9155	0.9648
Human	0.8912	0.9155	0.9032	0.9441	0.9648

Our approach Hybrid CNN, uses both RGB and ELA features to classify images as human or AI generated and reports a strong test accuracy of 92.43%. The ROC-AUC score of 0.9648 reports a strong discriminative power. F1 score for the AI label is 0.9506, for the human label 0.9032. The specificity values are good and indicate a balanced performance. The small performance difference between the two classes might be due to the class imbalance in the used dataset and the high realism of generated images. Nevertheless, the results are consistent and reliable.

4.2 EFFICIENTNET-B4 MODEL RESULTS

4.2.1 Training Performance

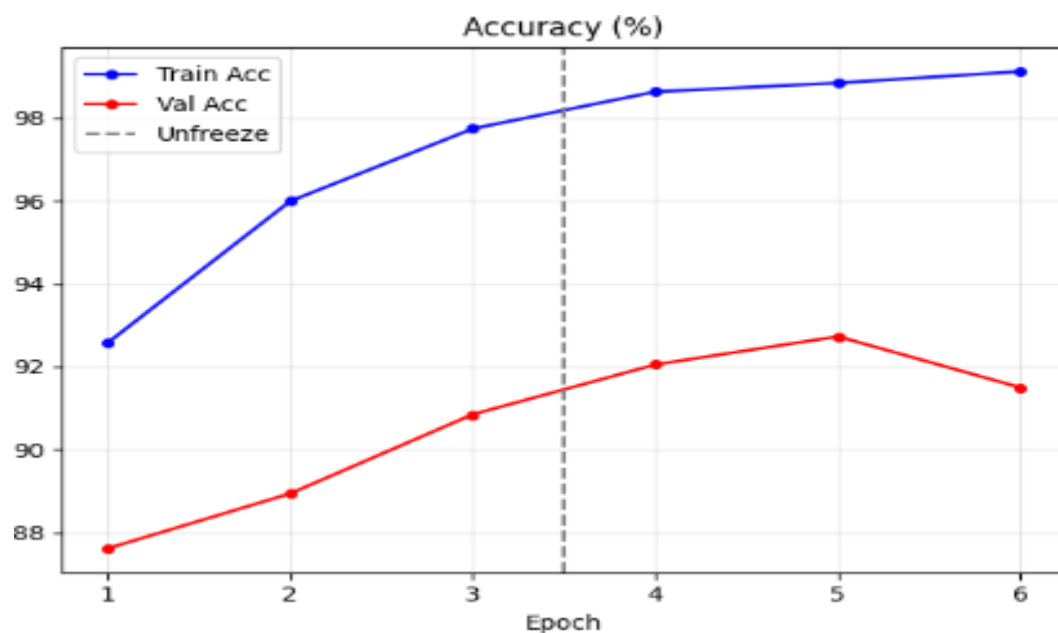


Fig. 4.4. Training and Test Accuracy Curves – Efficientnet-B4 Model

This is the EfficientNet-B4 model which was trained using transfer learning and then fine-tuned after it reached convergence. It reached about 99.1% training accuracy and 92–93% validation accuracy. The loss is consistent dropping for training and relatively flat for validation with slight overfitting after fine-tuning. Overall, the model demonstrates strong convergence and effective feature extraction capability.

4.2.2 Visualization of Evaluation Metrics

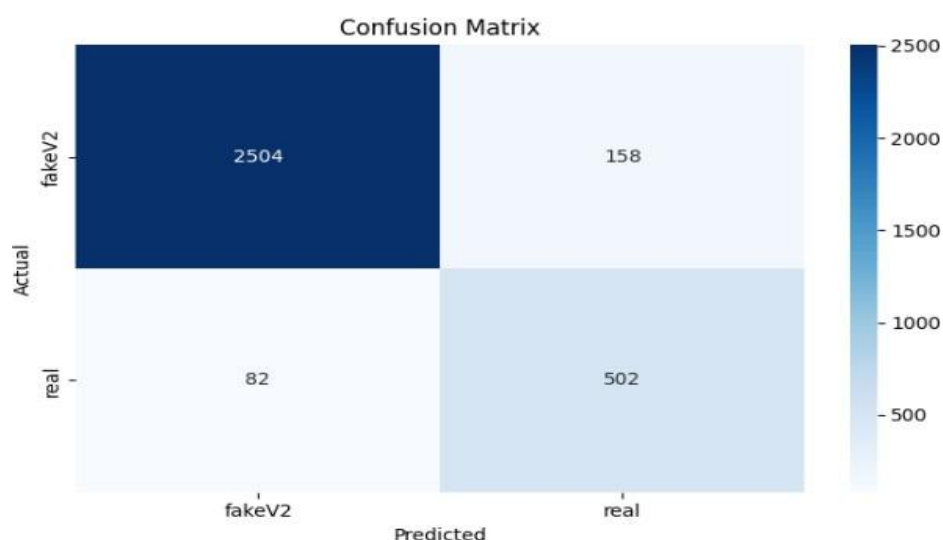


Fig. 4.5. Confusion Matrix – Efficientnet-B4 Model

The confusion matrix shows that the model correctly classifies the majority of AI images, with relatively few misclassifications. However, some real images are misclassified as AI, indicating slight bias toward the AI class.

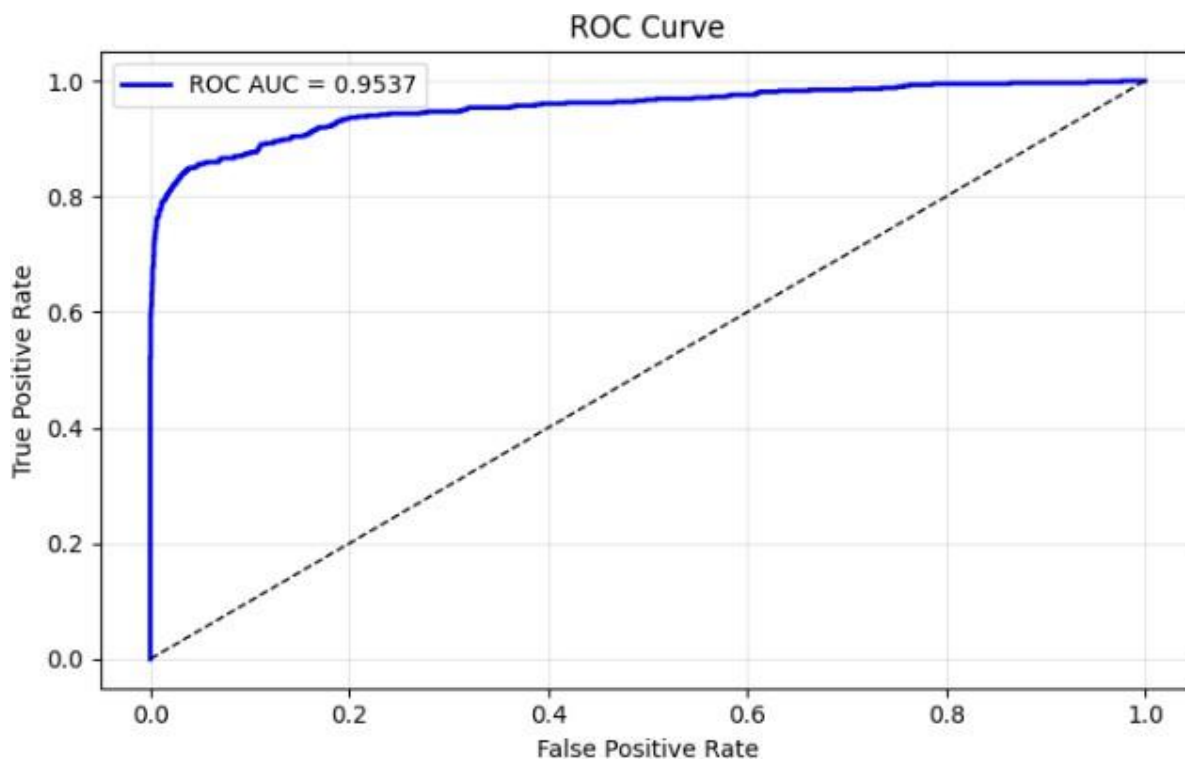


Fig. 4.6. ROC Curve – Efficientnet-B4 Model

Table 4.2 summarizes the general performance of the Efficientnet-B4 CNN model:

Table 4.2 Efficientnet-B4 CNN – Performance Metrics

Class	Precision	Recall	F1-Score	Specificity	ROC-AUC
AI	0.97	0.94	0.95	0.86	0.937
Human	0.76	0.86	0.81	0.94	0.937

The EfficientNet-B4 model obtained an accuracy of 93% with an overall ROC-AUC score of 0.9537. It specifically performed exceptionally well in the AI class with an F1 score of 0.95. However, its score in the human class was 0.81 which suggests it has room for improvement in real image detection. This can also be seen in the provided confusion matrix, where a considerable amount of real images were misclassified as being AI generated. This could be the result of the large class of AI images and the network's inherent bias towards this class due to dataset imbalance. Overall EfficientNet-B4 reached very high accuracy scores; however, they were not as evenly distributed as the scores reached by the Hybrid CNN model.

4.3 BASELINE CNN MODEL RESULTS

4.3.1 Training Performance

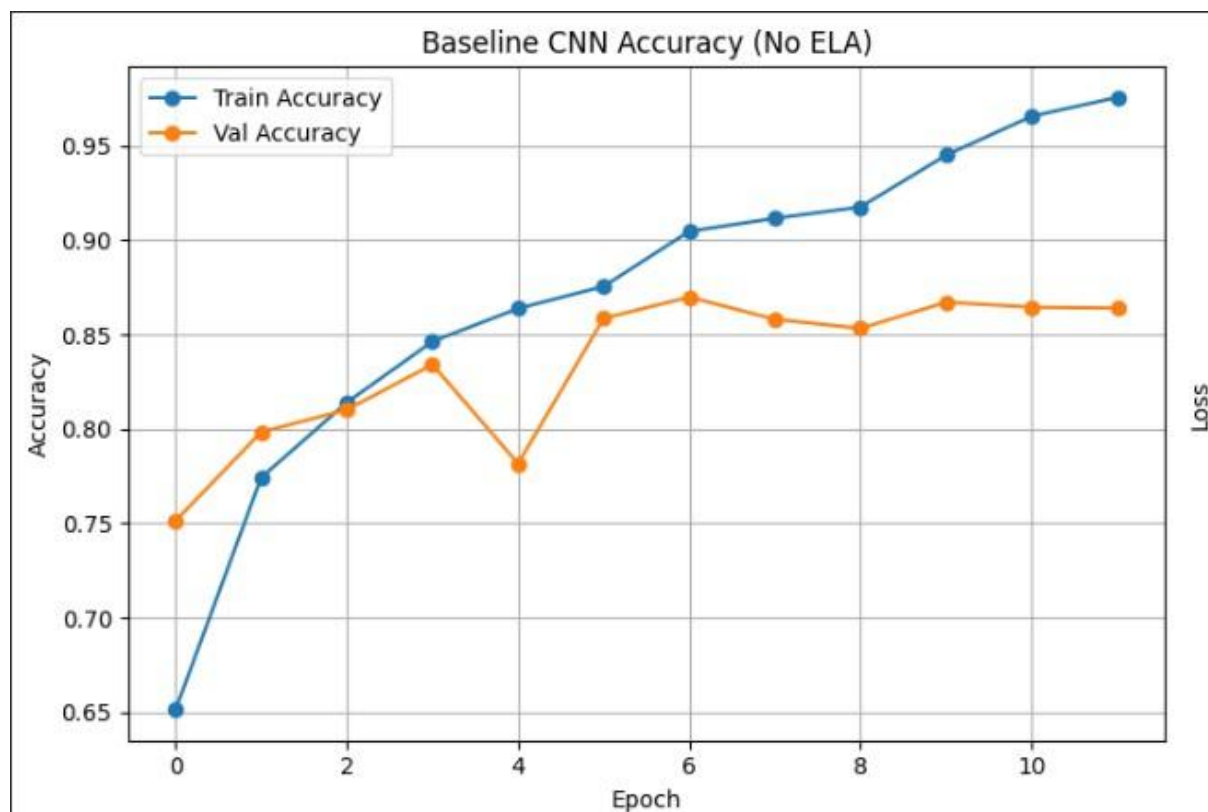


Fig. 4.7. Training and Test Accuracy Curves – Baseline CNN Model

We first trained the Baseline CNN on a reduced set of the images and did not utilise any ELA features. The training accuracy was steadily increasing from about 65% up to a correct classification rate of around 97%. On the other hand, the accuracy on the validation set only reached levels up to 86-87%. However, it did not reach a stable classification performance during the training run. Additionally, the loss values reveal overfitting as the training loss continued to decrease throughout the training, while the loss on the validation set varied and sometimes even increased during late epochs. This indicates that the model learned to merely

memorise the images from the training set rather than generalising also to images not seen before.

4.3.2 Visualization of Evaluation Metrics

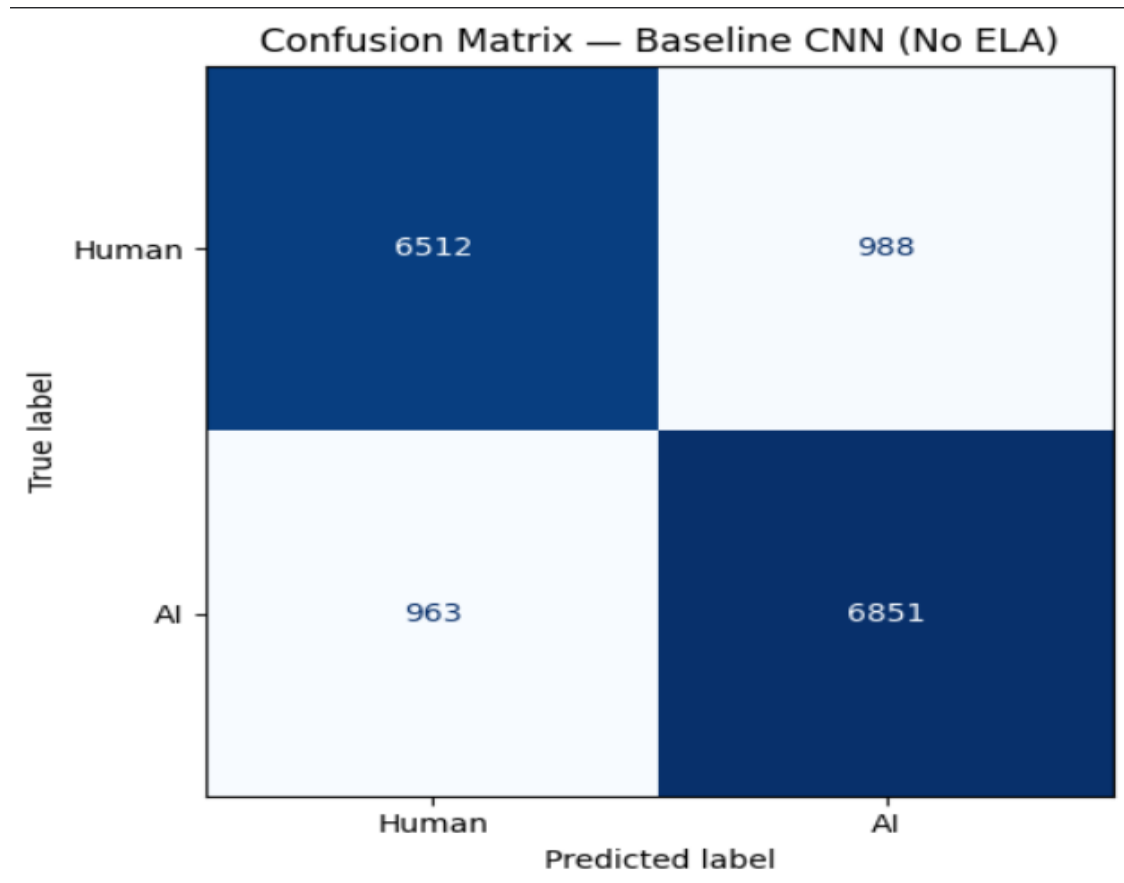


Fig. 4.8. Confusion Matrix – Baseline CNN Model

Confusion matrix indicates that the system performs reasonably well for both classes, but misclassifications are higher compared to the other models. Both False Positives and False Negatives are more prominent, reflecting limited discriminative capability.

Table 4.3 summarizes the general performance of the Baseline CNN model:

Table 4.3 Baseline CNN – Performance Metrics

Class	Precision	Recall	F1-Score	Specificity
Human	0.87	0.87	0.87	0.88
AI	0.87	0.88	0.88	0.87

4.4 COMPARATIVE ANALYSIS

Table 4.4 presents a comprehensive comparison of all three models implemented in this project. The comparison is based on overall test accuracy, average F1-score, and the computational approach used.

Model	Approach	Accuracy	Avg F1	Best Class F1
Hybrid CNN (RGB + ELA)	Deep Learning	92.43%	~0.927	0.9506 (AI)
EfficientNet-B4	Deep Learning (Transfer Learning)	93.00%	~0.880	0.95 (AI)
Baseline CNN	Deep Learning (Basic)	87.00%	~0.870	0.88 (AI)

Table 4.5 Comparative Analysis of All Four Models

4.5 EXPLAINABILITY ANALYSIS USING GRAD-CAM

Gradient weighted Class Activation Mapping (Grad-CAM) is used to enhance the interpretability of the proposed models. The technique assists in visualizing the parts of an image that make the largest contribution to the prediction of the model offering information on how the model decides.

Grad-CAM is used to create heatmaps by calculating the gradient of the target class from the final convolutional layer on the feature maps of that convolutional layer. These heatmaps highlight the most influential regions of the image that affect the output of the classification.

The visualization outcomes reveal that the model tends to focus on fine background artifacts, texture discontinuities, and compression patterns instead of the main object of the image. This is particularly evident with AI-generated images, in which artificial patterns and uneven distributions of pixels are identified.

On the contrary, in the case of human images, attention is more dispersed and naturally spread, which depicts naturalistic textures and consistency in structure. This is further improved with the incorporation of Error Level Analysis (ELA) which helps the model in detecting these artifacts and allows the model to capture those differences which cannot easily be detected by the human eye.

In general, the analysis using grad-cam proves that the model does not rely only object-level datas, but rather forensic-level features, which are effective in detecting AI-generated images. This enhances the transparency, interpretability and reliability of the system.



Fig. 4.9. Grad-CAM Visualization for AI-Generated Image

Grad-CAM determines parts of an image responsible for classifying a given image as AI-generated. The most responsible components are represented in warm colours (red and yellow) whereas the less responsible components are in cooler colours (blue).

From the illustrations, the model tends to ignore the focus of the image. Instead, the model tends to focus on areas with AI-generated images with discrepancies such as unnatural textures, inequitable lighting, and repetitive visual patterns.

The model, however, appears to ignore the focal object (the Character), but instead appears to work on fine background elements such as AI image artifacts. These background elements have fine compression artifacts and pixel disturbance artifacts that are utilized by the model, missing to the naked eye.

These findings suggest that the proposed Hybrid CNN model can learn more than just the model boundaries. Lastly, incorporating Grad-CAM provides evidence that the model does learn from AI image generation by utilizing important semantic cues, thus AI image generation model reliability and interpretability.



Fig. 4.10. Grad-CAM Visualization for Human (Real) Image

The Grad-CAM heatmap for the actual human image demonstrates the influence of image regions on the model's prediction. Red & yellow colors indicate important contributions, and blue and darker colors indicate lower contributions.

The focus of the model is given to the numerous natural features of the image such as different patterns of mountain texture, reflections of water, and lighting.

The model shows diffused attention over the number of natural elements of the image in the simplicity of the image. There are no artificial elements present in the image, which the Grad-CAM heatmap shows the model indicates.

CHAPTER 5

CONCLUSION AND FURTHER WORK

5.1 CONCLUSION

In our project, three deep learning models have been developed to differentiate between human generated images and those that have been created by a Generative Adversarial Network (GAN) and/or AI. A Hybrid CNN model using EfficientNet-B0 with Error Level Analysis (ELA) and a EfficientNet-B4 model using transfer learning on images of faces were built and tested. The results were then compared to a Baseline CNN that was trained from scratch to classify images that had been generated by AI. Through the use of feature fusion and more advanced deep learning architectures, the ability to accurately classify images has been tested and discussed.

The used dataset consists of real images and generated “artificial” images. For the real images we applied normalisation and ELA to enhance compression artefacts. The used models are a Hybrid CNN which uses both spatial features (RGB images) and forensic features (ELA images), EfficientNet-B4 which is using a pre-trained feature extraction, and a Baseline CNN as a comparison.

The results obtained from this project are summarised below.

- (a) The highest accuracy, 93%, was obtained by the EfficientNet-B4 model, which speaks to the efficacy of both transfer learning and deep feature extraction.
- (b) Hybrid CNN Model: By utilizing both RGB and ELA features, we obtained comparable accuracy of 92.43%, with more balanced classification rates.
- (c) For comparison, our simple Baseline CNN was able to achieve 87% accuracy, albeit lower than fine-tuned models. The model still showed some overfitting behavior, clearly demonstrating the value of effective feature extraction in regular neural network training.
- (d) Hybrid Approach: The Hybrid approach showed that using ELA-based forensic features improves significantly the robustness of a forensic-based model for detecting artifacts in images and videos generated by current state-of-the-art AI-based systems, and in particular for detecting very subtle artifacts.

- (e) EfficientNet-B4 learned to be biased towards the dominant class (AI images) whereas Hybrid CNN does not suffer from class-wise biased learning.

(f) Deep learning approaches significantly outperform basic CNN-based models. Furthermore, the utilization of domain specific preprocessing (ELA) significantly improves the reliability of classification results.

In this project, we studied the integration of deep learning techniques with existing methods of forensic image analysis, with a view to detecting AI-generated images. Generative models are becoming increasingly prevalent in computer graphics, and this work represents a feasible solution to detecting such images.

5.2 LIMITATIONS

Although the system is able to perform well on a number of tests, there are a few limitations with the current system.

- (a) The model performance is sensitive to image compression levels and preprocessing settings (especially ELA quality factor), which may affect consistency when applied to images from different sources.
- (b) The models are train, evaluated on the same datasets and performance is measured. The models are then tested on different datasets to gauge performance when the data has a different distribution.
- (c) The Baseline CNN was only trained on a subgroup of the dataset, which limits it's ability to generalize effectively.
- (d) The proposed system only deals with image-level detection and does not utilize metadata or context information.
- (e) ELA detection improves detection rates but is not perfect and some sophisticated manipulations may not be detected, for instance those generated by state of art deep learning based generative models.
- (f) Models require large computational resources to train, especially for models like EfficientNet-B4.

5.3 FURTHER WORK

This project work has informed possible future research directions:

Integration of Larger and Diverse Datasets: The output of this work could be used to train and test on a larger dataset. In addition to using images of people with PWS, models could be evaluated on images of people that have been generated by other forms of AI i.e. other types of deep learning-based image generation (e.g. other types of GANs, a diffusion model).

Advanced Feature Fusion Techniques: In the current stage, simple feature concatenation is employed. It may be worth exploring more sophisticated fusion of RGB and ELA features, such as attention-based fusion or more elaborate multi-modal learning approaches.

Real-Time Detection System: This model can be trained to work in a deep learning model and deployed as a real-time application or API that quickly identifies whether an image has been generated by AI, and can be very useful for both social media monitoring and cybersecurity use cases.

Improving Model Explainability: These methods can be improved to better explain deep learning decisions, for increased trust and interpretability. (e.g. Grad-CAM, Score-CAM)

Class Imbalance Handling: We would like to investigate the use of weighted-loss functions and data augmentation in future work.

Lightweight Model Development: As with many applications, developing optimized and lightweight models that can run on mobile or edge computing devices enables more practical usage of models for various use cases.

Multi-Modal Detection: Future versions of Multi-Modal Detection will not only focus on detecting highly realistic, state-of-the-art generated “deepfakes” by image features, but will also incorporate detection by text, metadata and source verification features.